

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd November 2020

Subject SP6 – Financial Derivatives

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Mark allocations are shown in brackets.***

Q. 1) The table below shows the current risk-free yield curve.

Term (years)	Risk Free Spot Rate
0.5	0.54%
1.0	0.61%
1.5	0.73%
2.0	0.85%
2.5	1.00%
3.0	1.15%
3.5	1.32%
4.0	1.48%
4.5	1.64%
5.0	1.80%
5.5	1.94%
6.0	2.08%

Note:

- All interest rates are expressed as continuously compounded per annum.
- All instruments considered below are payable semi-annually and the respective rates are convertible half-yearly.

i) Calculate the forward swap rate for a three-year swap starting at time 3. (3)

XYZ, a financial institution, is committed to paying an investor 3.5% p.a. interest (convertible half – yearly and payable semi-annually) on a Rs. 10 crores nominal amount for three years starting three years from now. The investor is likewise committed to depositing Rs. 10 crores with XYZ and earning 3.5% p.a. during that period.

XYZ wants to hedge this deal against interest rate falls. It is considering the following two options:

- Entering into a forward rate agreement (FRA) to receive 3.5% p.a. on the Rs. 10 crores nominal over the three – year period.
- Purchasing an at-the-money receiver swaption on a three-year swap with an exercise date in three years' time.

The Black volatility for such an at-the-money “three into three” swaption is 20% p.a.

ii) Calculate the price of the FRA. State whether XYZ would pay or receive this amount. (3)

iii) Calculate the price of the swaption. (5)

iv) Compare the two hedging approaches from a cost effectiveness point of view and risk point of view. (5)

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- Q. 2)** Let S_t denote the price of a non-dividend paying asset at time t , satisfying the stochastic equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where the drift μ and the volatility σ are both assumed constant and W_t is the Wiener process. Let C_t be the price of a derivative based on this asset satisfying the stochastic equation

$$dC_t = \mu_t^C C_t dt + \sigma_t^C C_t dW_t.$$

- i)** If at time “ t ” a trader is long φ_t units of the asset, and short one unit of the derivative, what is the value of trader position? Show that the position is risk free only if

$$\varphi_t = \frac{\sigma_t^C C_t}{\sigma S_t} \quad (3)$$

- ii)** If the short-term interest rate has a constant value, r , show that the principle of no arbitrage implies that

$$\frac{\mu_t^C - r}{\sigma_t^C} = \frac{\mu - r}{\sigma}.$$

Comment on the financial significance of this relationship. (3)

- iii)** If the derivative price can be expressed in the form $C_t = C(S, t)$ where the function $C(S, t)$ has a continuous second derivative. Use Ito's lemma to show that

$$C_t \mu_t^C = \frac{\partial C}{\partial t} + \mu S \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}$$

and

$$C_t \sigma_t^C = \sigma S \frac{\partial C}{\partial S}. \quad (4)$$

- iv)** Show that the no arbitrage condition implies that $C(S, t)$ satisfies

$$\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = r \left(C - S \frac{\partial C}{\partial S} \right).$$

What is the significance of the parameter μ that does not appear in this relation? (4)

[14]

- Q. 3)** Answer the following questions pertaining to barrier options

- i)** Describe the properties of barrier options. (1)

- ii)** Explain why an investor would consider using barrier options. (1)

iii) Determine a portfolio of barrier options which is equivalent to a European plain vanilla call option. Ignore transaction costs. (1)

iv) A bank has purchased a one-year down-and-out barrier put option which is European plain vanilla on expiry, but which has an American barrier.

a) Describe the possible pay-outs from this option during its lifetime. (2)

b) This barrier option is written on the AUD/GBP exchange rate. The strike price is 0.77 AUD and the barrier are 0.70 AUD. The nominal is 20 million AUD and the remaining time to expiry is one month. Explain how to construct a graph of the delta of this barrier option against the exchange rate at this duration taking in to account the different zones the exchange rate may land up in. (3)

c) The bank decides to delta hedge its position. Discuss the problems the bank may encounter in maintaining this hedge, including the effect on currency markets. (2)

[10]

Q. 4) A financial institution is considering the use of Vasicek model of interest rates

i) Describe the Vasicek one-factor model of interest rates and its key statistical properties. (2)

ii) Let r_t denote the short rate of interest applicable at time t . According to a parameterization of this model, the instantaneous forward rate applicable at time $t > 0$ implied by the market prices at time 0 is found to be:

$$f(t) = r_0 e^{-\alpha t} + r_\infty (1 - e^{-\alpha t}) + k(1 - e^{-\alpha t})e^{-\alpha t}$$

where, α is a positive constant.

a) Show that, if the graph of $f(t)$ has a stationary point, then

$$e^{-\alpha t} = \frac{r_0 - r_\infty + k}{2k} \quad (2)$$

b) Hence describe the shape of the graph of $f(t)$ in each of the following cases

i. $r_0 = 0.04, r_\infty = 0.06, k = 0.1$

ii. $r_0 = 0.06, r_\infty = 0.04, k = 0.02$ (2)

c) State a possible explanation for a yield curve with the shape of the graph in (ii)(b)(i) (2)

iii) Describe the main advantages and limitations of the Vasicek model. (2)

[10]

Q. 5) An insurance company is planning to enter into an uncollateralised swap agreement as a part of its hedging activities.

- i) Explain the difference between a collateralised swap and an uncollateralised swap and the risk exposure in each case. List the types and timing of collaterals used by banks in such transactions. (3)
- ii) The insurance company enters into an uncollateralised swap with a company Broch infrastructure (a road building company in the infrastructure sector) at par to exchange 5% fixed with floating for 4 years to be paid annually at the end of each year based on floating rate for next 12 months. (i.e. 4th payment would cover time between year 4 to 5). The nominal of the transaction is 100cr. After 12 months (just before next payment) interest rate reduced by 1%. Estimate the value of the swap for the company after 12 months just before the 1st payment is made. All rates are annually compounding. (3)
- iii) New regulation has stipulated that all the insurance companies should set aside 300% of expected loss due to uncollateralised asset as capital requirement. As an investment actuary, you have been requested to set out the approach to assess the capital requirement due the swap and estimate the value based on the below information provided by a market data provider.

Year	Number of swaps (across all sectors)	Defaults on payment
2015	890	43
2016	740	24
2017	1010	71
2018	780	28
2019	790	67

Average recovery of the defaulted swaps has been historically 30%. State all the assumptions used in the calculation. (8)

- iv) Peer review actuary doing the validation of your capital calculation has suggested using corporate bonds of Broch to assess expected loss from uncollateralised swap instead of relying on sector data. Critique the observation highlighting its strengths and weaknesses. Suggest approaches to account/hedge for the residual risk if corporate bond approach is used. (5)
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Q. 6) XYZ Corp is a buyer of crude oil and wants to hedge its exposure using crude oil futures

- i) What is minimum variance hedge ratio and where can this be used? (2)
- ii) XYZ Corp has a requirement for 10,000 barrels of oil. Volatility of spot and future returns has been 4% and 6% respectively in the past and the correlation between their returns is 75%. Suggest the number of future contracts XYZ Corp should buy that would minimise the risk associated with an unexpected increase in the price of oil. Each contract is for 1000 barrels. (2)

- iii) Discuss any residual risk which XYZ Corp would be exposed to due to this transaction.

(2)

[6]

Q. 7) An Insurance company has the following ALM split with exposure of 1000Cr.

	Current
Bonds Gov	50%
Bonds Corp	30%
Indian equity	10%
US equity	10%

The current regulation limits equity exposure to 20%. Regulator is suggesting new regulation allows increasing the asset allocation of equities from 20% to 50% and the ALM team has suggested the new asset allocation

	Future
Govt. Bonds	Min 25%
Corp. Bonds	Max 25% and min 10%
Indian equity	30%
US equity	20%

As an investment actuary you have to make a recommendation on decision to implement the above changes over the period of 6 months

- i) Current market expectation is Government bond yield to fall and along side increased default in corporate bond. Suggest what proportion of Government bonds and corporate bonds be sold to attain the future asset allocation. (2)
- ii) Given the market uncertainty it's been recommended to use futures to carry out the following asset allocation shift. State advantage of using futures compared to direct sale of asset in the market. Highlight the challenges of using futures for bonds. (4)
- iii) Suggest the transaction the company should take to achieve the above asset allocation along with value of future transaction. Please use the following information
 - There are no futures available for corporate bonds
 - Spot FX rate – USD/INR - 75
 - S&P 500 index today 3000, BSE index today 35000 [assume this to be price for the single future contract]
 - We expect 6% annual coupon payment from Government bond over next 6 months. Each unit for future for Government bond is 1cr

- Interest rate India 6% per annum half yearly compounded and US 2% per annum half yearly compounded (6)

- iv) The company is planning to issue a perpetual bond with 5% coupon payment (which is equal to expected bond yield) annually with Principle of 1 Cr. Derive formula for estimating the price, and convexity of the bond. (5)

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Q. 8) Recently the price of crude oil futures in US slipped into the negative territory.

- i) What are the implications of this event on entities using Black Scholes option pricing model and how they can respond to it? (5)

- ii) An oil trading desk uses an artificial intelligence powered predictive model that predicts the crude oil price based on global inventory levels. Describe the relationship between crude oil price and inventory. What kind of changes will this model undergo now? (3)

[8]
